

Project Title

AI-Powered Refugee Population Forecasting and Humanitarian Resource Planning System for Kenya

Framework

CRISP-DM

Data Domain

Humanitarian Analytics, Migration Studies, Population Forecasting, Public Policy

Project Objective

Kenya hosts refugee and asylum-seeking populations from multiple countries affected by conflict, political instability, economic crises, and natural disasters.

Humanitarian agencies often face challenges in forecasting refugee population changes, resulting in resource shortages or inefficient allocation of aid.

This project aims to develop a predictive analytics and deep learning system capable of forecasting refugee population trends across demographic groups and supporting evidence-based humanitarian planning.

1. Business Understanding

Problem Statement

Refugee populations are dynamic and can change rapidly due to:

- Armed conflicts
- Political instability
- Economic crises
- Climate-related disasters

Humanitarian organizations need predictive insights to:

- Anticipate population growth
- Allocate resources efficiently
- Plan healthcare services

- Improve educational support
- Enhance shelter planning

Current reporting is largely retrospective rather than predictive.

2. Data Understanding

Dataset

Kenya Refugee and Asylum Population Dataset

Dataset Size

- 27,664 observations
- 14 variables

Features

Feature	Description
origin_location_code	Country of origin
origin_has_hrp	Humanitarian Response Plan indicator
origin_in_gho	Included in Global Humanitarian Overview
asylum_location_code	Host country
asylum_has_hrp	Host HRP indicator
asylum_in_gho	Host GHO indicator
population_group	Refugee/Asylum category
gender	Male/Female
age_range	Age category
min_age	Lower age bound
max_age	Upper age bound
population	Population count

Feature	Description
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reference_period_start	Start period
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reference_period_end	End period
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Target Variable

Population

The total refugee/asylum population count.

3. Data Preparation

Data Cleaning

- Handle missing values
- Remove inconsistencies
- Validate age ranges
- Check population anomalies

Feature Engineering

Time Features

Extract:

- Year
- Quarter
- Month

Demographic Features

Generate:

- Working-age population
- Child dependency ratios
- Elderly population indicators

Geographic Features

Encode:

- Origin country
 - Regional groupings
 - Conflict-prone regions
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4. Exploratory Data Analysis

Key questions:

- Which countries contribute the largest refugee populations?
- What are the demographic distributions?
- How have refugee populations changed over time?
- Which age groups dominate refugee movements?
- Are there gender disparities?

Visualizations:

- Population pyramids
 - Trend analysis
 - Geographic flow maps
 - Time-series dashboards
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5. Machine Learning

Population Forecasting

Models

- Linear Regression
- Random Forest Regressor
- XGBoost Regressor
- LightGBM

Goal

Predict future refugee population counts.

Classification Model

Objective

Predict whether a refugee population segment will experience significant growth.

Target:

0 = Stable Population

1 = Rapid Population Increase

Models

- Logistic Regression
 - Random Forest
 - XGBoost
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6. Deep Learning

LSTM Forecasting Model

Purpose:

Forecast future refugee population trends using historical population sequences.

Benefits:

- Captures temporal patterns
 - Detects long-term migration trends
 - Supports proactive humanitarian planning
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Neural Network Prediction Model

Predict:

- Population growth
- Demographic shifts

- Future resource requirements
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7. Resource Planning Analytics

Using predicted populations, estimate future needs for:

- Shelter capacity
- Food assistance
- Education facilities
- Healthcare services

This transforms the project from simple forecasting into decision-support analytics.

8. Deployment

Solution Architecture

Historical Data

→ ETL Pipeline

→ ML/DL Models

→ API Layer

→ Dashboard

Tools

- Python
 - FastAPI
 - Streamlit
 - Docker
 - AWS or Render
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Dashboard Features

Users can:

- View refugee trends

- Forecast future populations
 - Analyze demographic breakdowns
 - Compare countries of origin
 - Monitor high-growth refugee groups
 - Estimate humanitarian resource requirements
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Expected Outcomes

1. Refugee population forecasting model.
 2. Deep learning migration prediction system.
 3. Humanitarian resource planning dashboard.
 4. Deployable AI solution for decision support.
 5. Early warning system for refugee population growth.
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Business Value

The project enables humanitarian agencies and policymakers to transition from reactive planning to predictive planning by anticipating refugee population changes and allocating resources more effectively.